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OPERATING SYSTEM CONCEPTS LAB EXPERIMENTS

EXP – 05: SCHEDULING ALGORITHMS (Shortest Job First)

Activities Terminal Mar 25 00:44

user123@localhost:~/scheduling

```
[user123@localhost ~]$ ls
abc.sh  Desktop    Downloads  ifelse.sh  Music  Pictures   Public   REC       sample.sh   scheduling  sh         Videos
ab.sh   Documents  Folder     loops.sh   pc     posneg.sh  qns.txt  sample.c  sample.txt  SFA         Templates  while.sh
[user123@localhost ~]$ cd scheduling
[user123@localhost scheduling]$ vi sjf.c
[user123@localhost scheduling]$ cat sjf.c
#include <stdio.h>

int main(){
        int n;
        printf("Enter number of processes: ");
        scanf("%d",&n);

        int at[n], bt[n], ct[n], tat[n], wt[n];
        int done[n];

        for(int i=0; i<n; i++){
                printf("Enter Arrival Time and Burst Time for P%d: ",i+1);
                scanf("%d %d",&at[i], &bt[i]);
                done[i] = 0;
        }

        int completed = 0;
        int completionTime = 0;
        float avgtat = 0, avgwt = 0;

        while(completed < n){
                int idx = -1;
                int minBT = 9999;

                for(int i=0; i<n; i++){
                        if(at[i]<=completionTime && done[i] == 0){
                                if(bt[i]<minBT){
                                        minBT = bt[i];
                                        idx = i;
                                }
                        }
                }

                if(idx != -1){
                        if(completionTime<at[idx]){
                                completionTime = at[idx];
                        }
                        completionTime += bt[idx];

                        tat[idx] = completionTime - at[idx];
                        wt[idx] = tat[idx] - bt[idx];

                        done[idx] = 1;
                        completed++;
                } else {
                        completionTime++;
                }
        }

        for(int i=0; i<n; i++){
                avgtat += tat[i];
                avgwt += wt[i];
        }

        printf("\nAverage Turnaround Time: %.2f\n",avgtat/n);
        printf("Average Waiting Time: %.2f\n",avgwt/n);

        return 0;
}
[user123@localhost scheduling]$
```

```
[user123@localhost scheduling]$ gcc sjf.c
[user123@localhost scheduling]$ ./a.out
Enter number of processes: 4
Enter Arrival Time and Burst Time for P1: 1 3
Enter Arrival Time and Burst Time for P2: 2 4
Enter Arrival Time and Burst Time for P3: 1 2
Enter Arrival Time and Burst Time for P4: 4 4

Average Turnaround Time: 6.25
Average Waiting Time: 3.00
[user123@localhost scheduling]$ [redacted]
```

SFA - 241401113